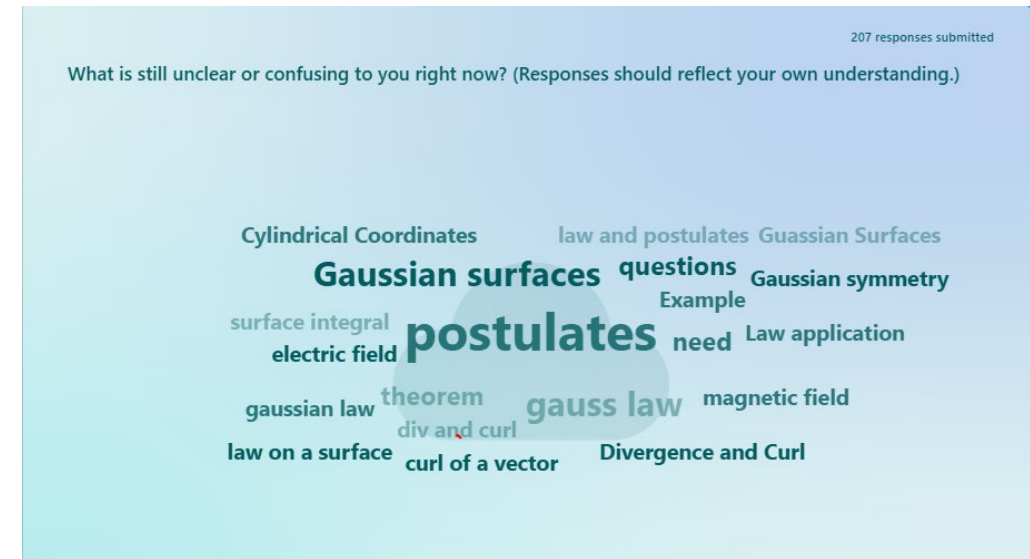


ELEC 3115

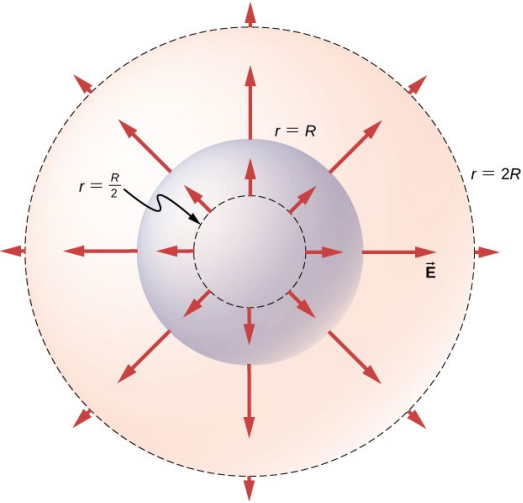
Lecture 2: Week 1 , Thursday

Your responses at the end of lecture 1:



Review

Let's observe the $\mathbf{E}$ of a spherical charged body (like the charged cloud of Example 1):



How would you characterise the flux lines (shown by the red arrows) – diverging/converging/circulating around the centre point?

If the volume charge density ρ is at the centre point,

Postulate 1: $\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon}$

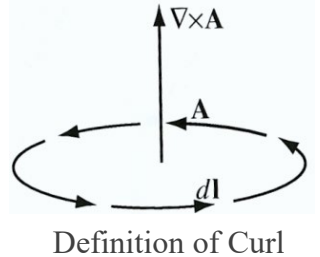
Divergence theorem

$$\oint_S \vec{E} \cdot d\vec{S} = \frac{Q_{enclosed}}{\epsilon} \quad (\text{Gauss's law})$$

Postulate 2: $\vec{\nabla} \times \vec{E} = 0$

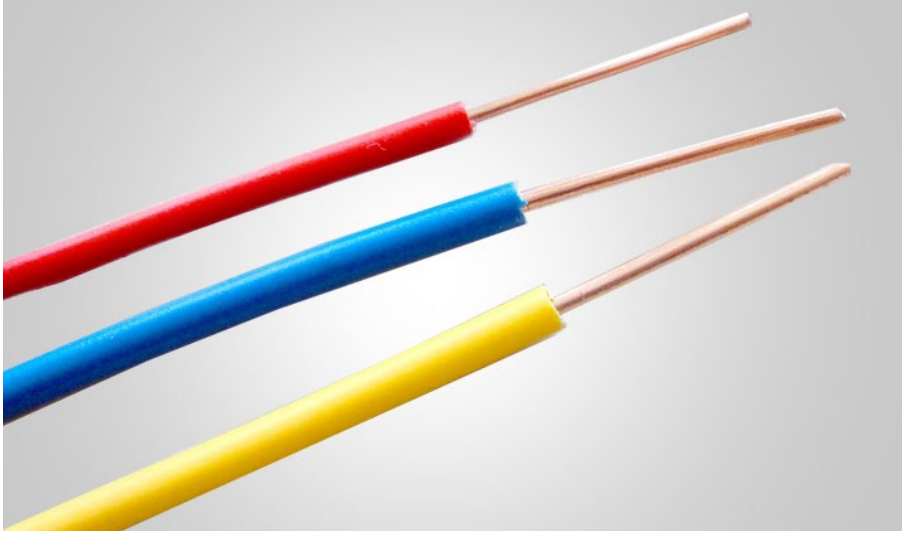
Stokes theorem

$$\oint_C \vec{E} \cdot d\vec{l} = 0 \quad (\text{KVL})$$



- A **Gaussian surface** can be found if we can determine symmetry (i.e., if we can match the geometry of the charged object to an object with surfaces on which $\mathbf{E}$ is normal)
- **Helmholtz's theorem:** if a vector is not circulating, it must be increasing/decreasing in some direction. Hence, it must be a gradient of a scalar.

$$\vec{E} = -\vec{\nabla} V \rightarrow V_{ba} = -\int_a^b \vec{E} \cdot d\vec{l}$$
- What has happened to the **magnetic field**? A magnetic field will appear when we consider charge is not stationary.



Today, we will try to find answers:

A transmission line carries charges.

- What is the electric field inside the conductor?
- What is the potential at the surface?
- How much voltage exists between the line and ground?
- Will the insulation survive?

To answer these, we must move from **$E \rightarrow V$**
and from **vacuum $\rightarrow$ real materials**

Today's learning goals

By the end of today's lecture, you will be able to:

1

Compute the electric potential V from a known electric field E .
(in cylindrical geometries such as transmission lines)

2

Explain why the electric field inside a conductor is zero
and describe the behaviour of its normal and tangential components at the surface.

3

Describe how dielectric materials respond to electric fields
using the concepts of polarization and relative permittivity (ϵ_r)

4

Relate dielectric strength to electric field limits
in real engineering systems (e.g., insulation breakdown).

Electric Field and Potential of a Cylindrical Conductor

Why this matters for engineers:

Power cables and transmission lines behave approximately like long charged cylinders.

In this section, you will:

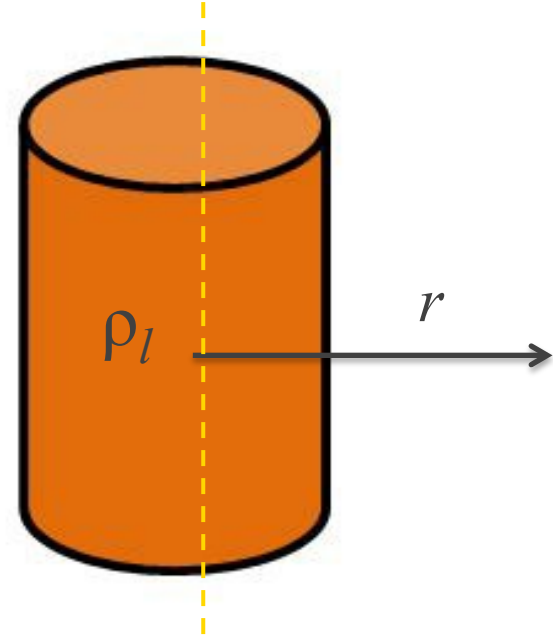
- Use Gauss' law to derive $E(r)$
- Integrate to obtain $V(r)$
- Understand why the potential varies logarithmically

Potential differences around a charged cylindrical body

Electrical field at radial distance r from the centre of a cylindrical body with line charge density ρ_l [C/m]:

$$\oint_s \vec{E} \cdot d\vec{s} = \frac{Q_{enclosed}}{\epsilon_0} \rightarrow 2\pi r L |E| \vec{a}_R = \frac{\rho_l L}{\epsilon_0} \Rightarrow \vec{E} = \frac{\rho_l}{2\pi\epsilon_0 r} \vec{a}_r$$

(like example 2)



Potential difference between two radial points:

$$V_{21} = -\int_{r_1}^{r_2} \vec{E} \cdot d\vec{l} = -\int_{r_1}^{r_2} \frac{\rho_l}{2\pi\epsilon_0 r} dr = \frac{\rho_l}{2\pi\epsilon_0} \ln \left(\frac{r_1}{r_2} \right)$$

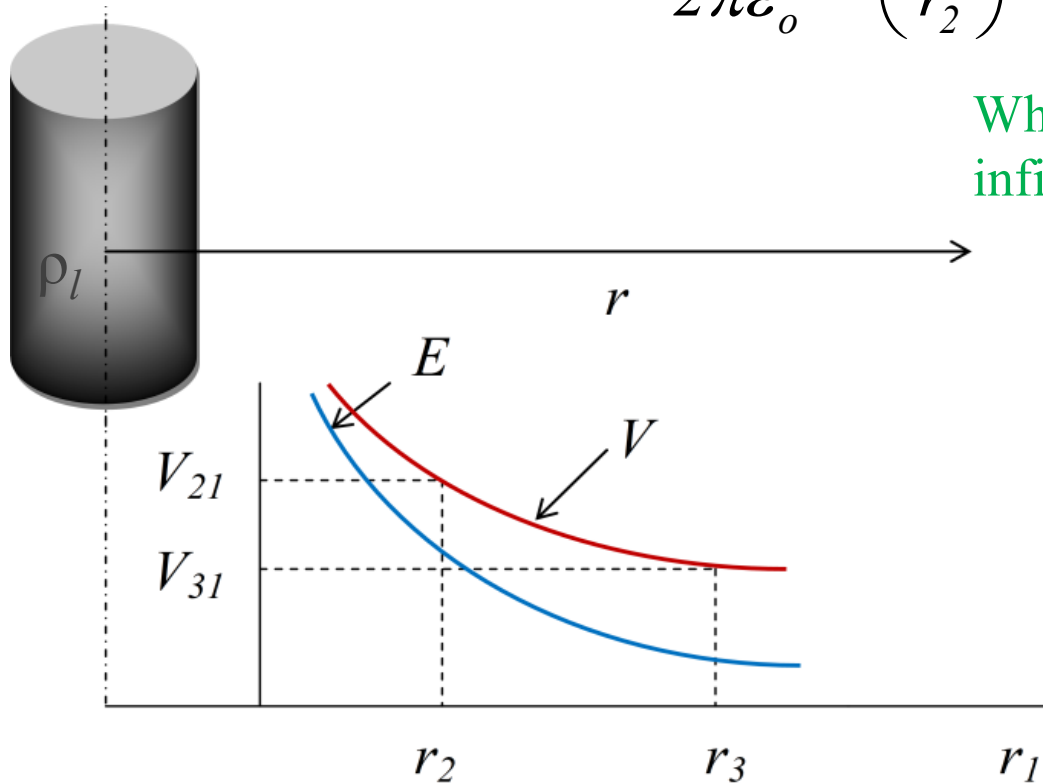
Why will we use this potential function $V(r)$?

- Estimate insulation stress
- Predict breakdown
- Design safe insulation thickness

This is not “math practice” — this is cable design.

Similarly,
$$V_{3l} = - \int_{r_1}^{r_3} \vec{E} \cdot d\vec{l} = - \int_{r_1}^{r_3} \left(\frac{\rho_l}{2\pi\epsilon_0 r} \right) dr = \frac{\rho_l}{2\pi\epsilon_0} \ln \left(\frac{r_1}{r_3} \right)$$

$$\therefore V_{23} = V_{2l} - V_{3l} = \frac{\rho_l}{2\pi\epsilon_0} \ln \left(\frac{r_1}{r_2} \right) - \frac{\rho_l}{2\pi\epsilon_0} \ln \left(\frac{r_1}{r_3} \right) = \frac{\rho_l}{2\pi\epsilon_0} \ln \left(\frac{r_3}{r_2} \right)$$



Variation of $\mathbf{E}$ and V with r

Why does V vary logarithmically? Can potential be zero at infinity for a line charge density?

What You Must Remember:

$$\vec{E}(r) = \frac{C}{r} \vec{a}_r \rightarrow V(r) = C \ln(r), \text{ where } C = \frac{\rho_l}{2\pi\epsilon}$$

- ✓ You are expected to derive $V(r)$ using Gauss's law.
- ✓ You are NOT expected to memorise the logarithm form without understanding.

Example 3

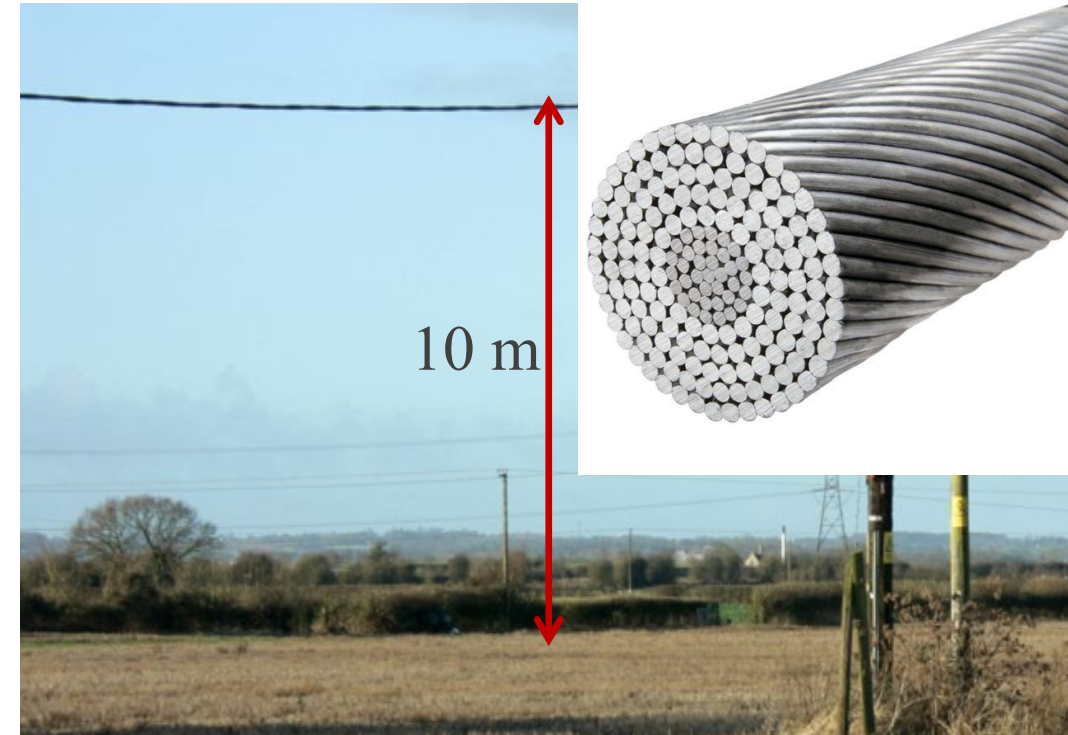
If in Example 2 (ii), diameter of the transmission line is 2 cm, what is the potential difference at the surface of the line with respect to a reference located at ground below at 10m.

Ans: 12.4kV $\epsilon_o = 8.854 \times 10^{-12} \text{ F/m}$, $\rho_l = 10^{-7} \text{ C/m}$

Solution hints:

Use GL to find $\mathbf{E} \rightarrow V \rightarrow$ determine $r_1, r_2 \rightarrow$ calculation

$$\oint_s \vec{E} \cdot d\vec{s} = \frac{Q_{\text{enclosed}}}{\epsilon_0} \rightarrow \vec{E} = \frac{\rho_l}{2\pi\epsilon_0 r} \vec{a}_r \rightarrow V_{21} = -\int_{r_1}^{r_2} \vec{E} \cdot d\vec{l} \rightarrow V_{21} = \frac{\rho_l}{2\pi\epsilon_0} \ln \left(\frac{r_1}{r_2} \right)$$



Conductors in Electrostatics

Core principle:

Free charges move until the internal electric field becomes zero.

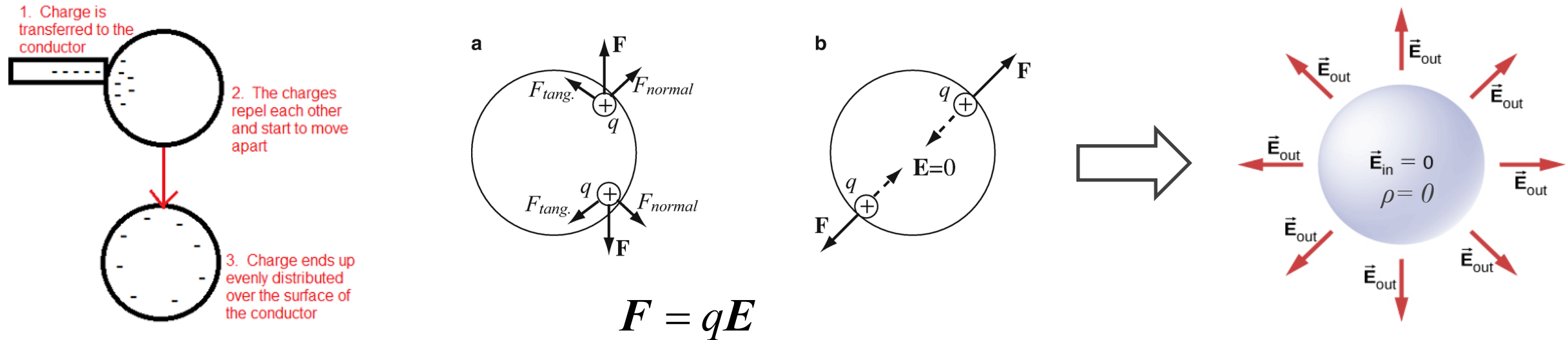
In this section, you will:

- Explain why $E = 0$ inside a conductor
- Distinguish between normal and tangential components of E
- Understand surface charge formation

Conductors and electric field

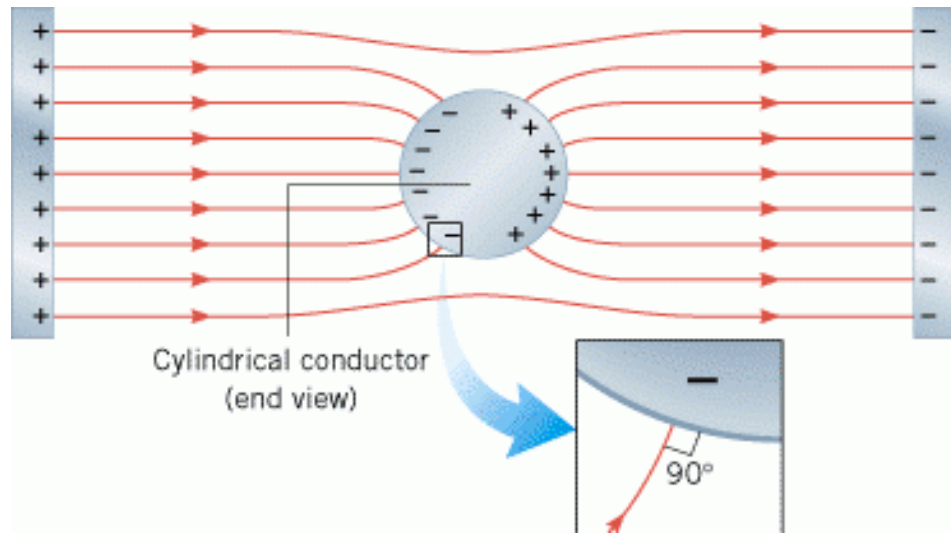
Outer electrons are loosely bound to their atoms and can move freely. Hence, it a very good career of electricity.

Charging of an isolated conductor

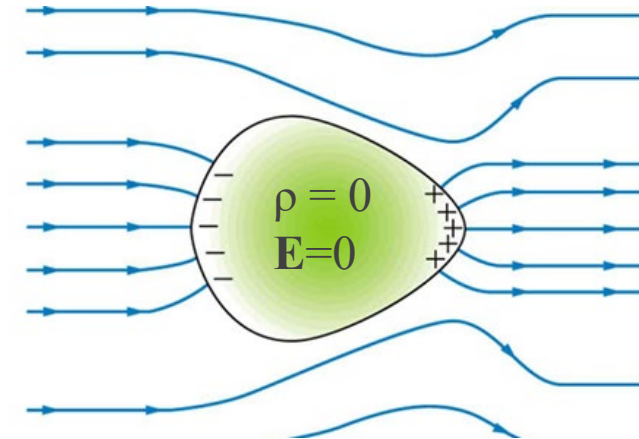
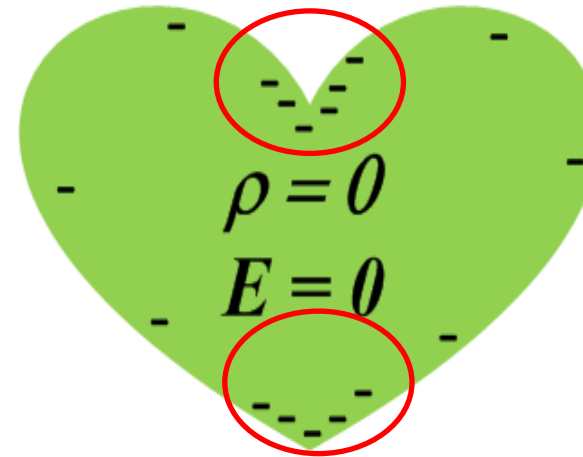


Repelling force (i.e. the tangential component of the force) pushes the charges of the same polarity to the surface until it reaches an equilibrium (i.e., zero net force). Thus, no $\mathbf{E}_{tang}$, only $\mathbf{E}_{normal}$ exists. No ρ and $\mathbf{E}$ inside. Charges are on the surface: ρ_s exists, for a long conductor, it is ρ_l .

Isolated conductor in an external electric field

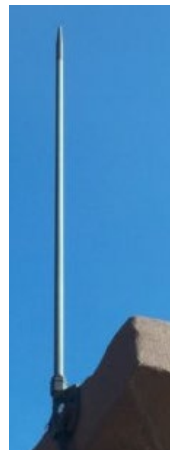


Charges at uneven conductor surface



Note how the charges crowd at the sharp edges. E outside is perpendicular to the surface outside.

Lightening Arresters:



E and V of a charged conductor: summary

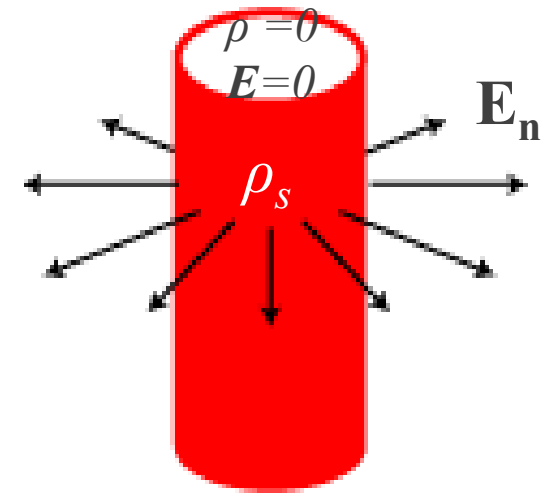
- ρ [C/m³] inside a charged (whole) conductor is zero; but ρ_s [C/m²] or ρ_l [C/m] exists.
- No **E** or **D** inside the conductor. **E** or **D** normal to the outer surface exist outside, but no **E** or **D** tangential to the surface.

$$\left. \begin{aligned} \vec{E}_n &= \frac{\rho_s}{\epsilon_0} \vec{a}_n, \vec{D}_n = \rho_s \vec{a}_n, \\ \vec{E}_n &= \frac{\rho_l}{2\pi\epsilon_0 r} \vec{a}_n, \vec{D}_n = \frac{\rho_l}{2\pi r} \vec{a}_n \end{aligned} \right| \vec{E}_t = 0, \vec{D}_t = 0$$

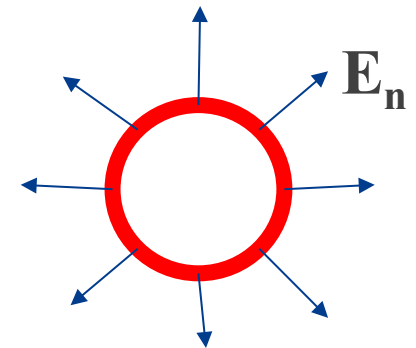
- *Potential at the surface with respect to a reference at r_o :*

$$V = \frac{\rho_l}{2\pi\epsilon_0} \ln \left(\frac{r_o}{r} \right) \quad \text{The outside surface is at equipotential.}$$

r : radius of the conductor.



Electric field in a cylindrical conductor



Top view

Pause & Reflect

(Before we take a short break...)

 **Take 1 minute to jot down:**

One idea you just learned

One step you could explain to a friend

One word that describes how confident you feel right now
(scale: 😞 *unsure* → 😊 *okay* → 😄 *confident*)

💡 *This is for you — no submission.*



Welcome back — anyone happy to share a key idea they jot down?

Dielectrics & Polarization

What Happens Inside an Insulator?

Key question:

If charges cannot move freely, how does a material respond to an electric field?

In this section, you will:

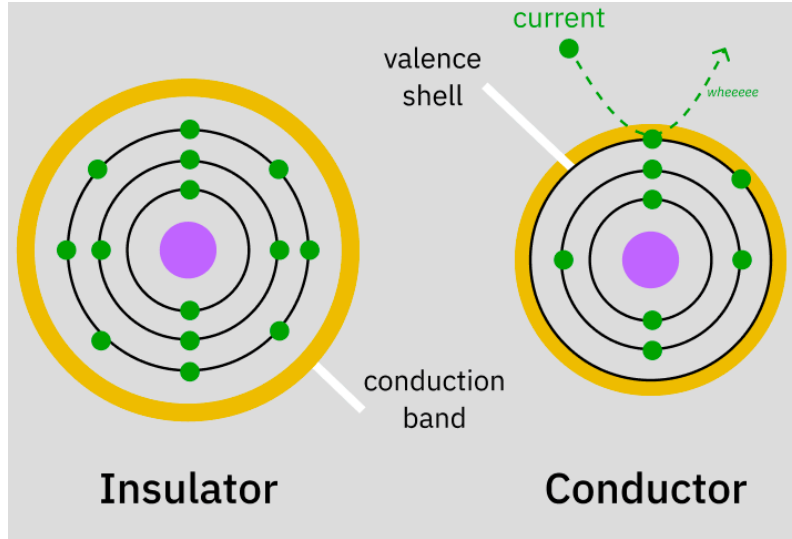
Understand dipoles and polarization

Define the polarization vector $\mathbf{P}$

Interpret bound charge physically

Insulators or Dielectrics

-No free electrons, and bad conductor of electricity.



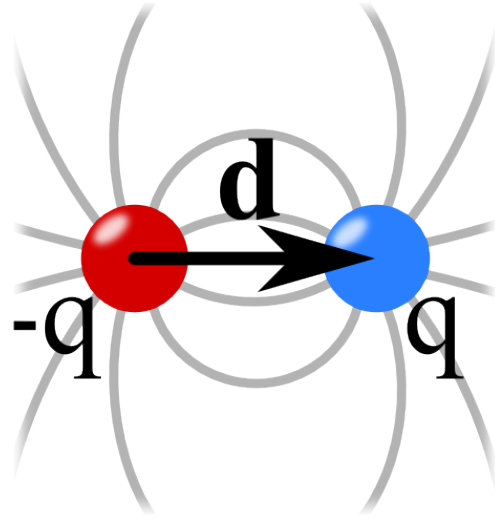
Examples :
polyethylene, mica, rubber, paper, silica.

When dielectrics are subjected to an external electric field, dipoles develop at microscopic level.

What is a dipole? How do you measure a dipole?

What are electric dipoles?

Two equal but opposite charges separated by a short distance.



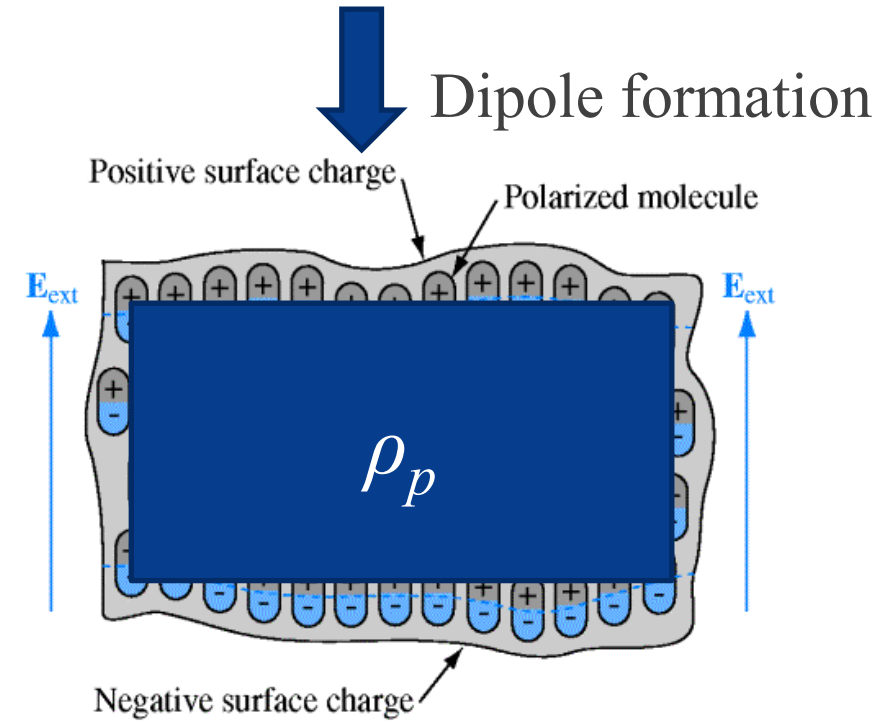
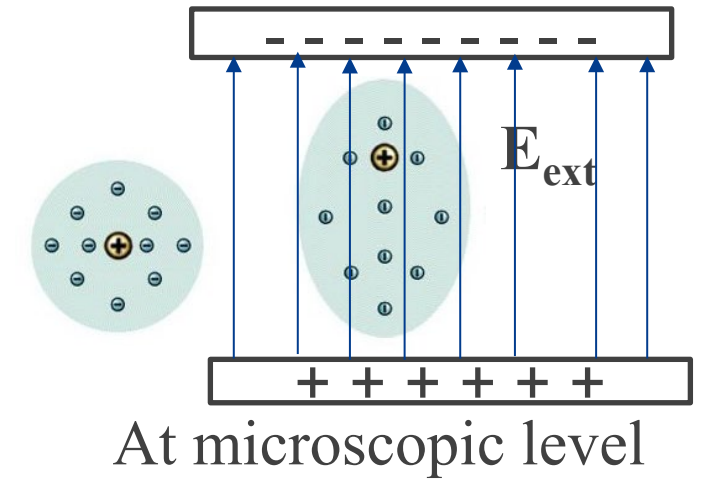
Dipole moment: $\vec{p} = q\vec{d}$ What is the unit of dipole moment?

Effect of electric field in dielectric materials

The process of dipole formation in dielectrics due to an external electric field is known as **polarization**.

Polarization is equivalent to inducing a *polarization charge density* or *bound-charge density* ρ_p in the volume of a dielectric.

Note that the net charge inside a dielectric after polarization is zero (i.e., the dielectric has an equal amount of positive and negative charges).



The Polarization Vector

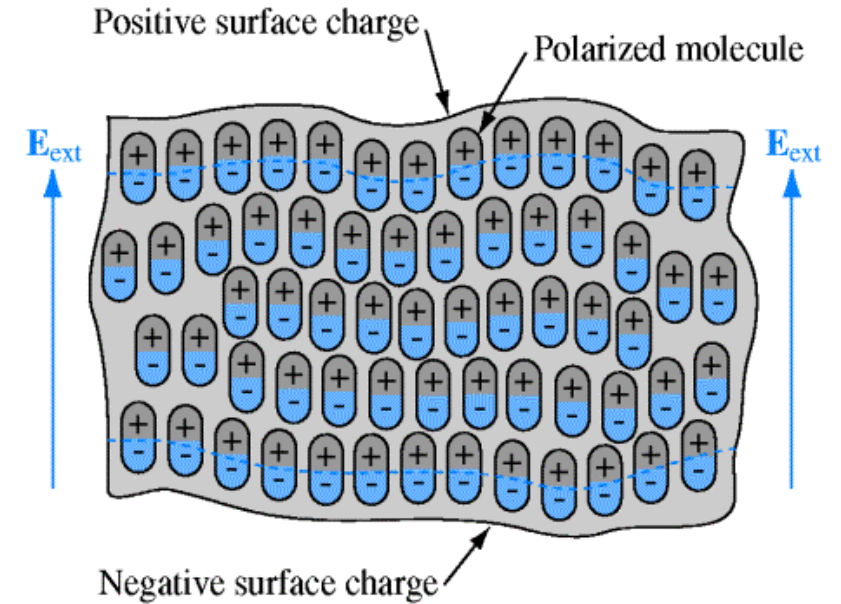
To measure the microscopic effect of dipole formation, we introduce a polarization vector.

Number of dipole moments per unit volume of a dielectric is the **polarization vector** (i.e., the volume density of dipole moments):

$$\vec{P} = \lim_{\Delta v \rightarrow 0} \frac{\sum_{k=1}^{n\Delta v} \vec{p}_k}{\Delta v} \quad (\text{C/m}^2)$$

where, $\vec{p}_k = q\vec{d}$ is a dipole moment (unit C-m)

n is the number of dipoles in a small volume Δv



Note the unit of Polarization vector! It is the surface charge density of the bound charges.

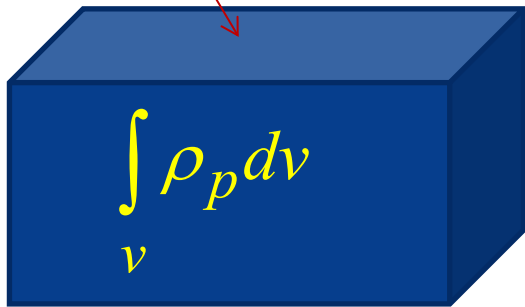
How polarization vector $\mathbf{P}$ relates to bound charge density ρ_p

Closed surface integration of $\mathbf{P}$ (since it is the surface charge density) gives the total bound charges in the closed surface of the dielectric due to polarization.

Volume integration of ρ_p gives the total bound charges that remain in the volume.

Since the net charge in the dielectric is zero,

$$\oint_S \vec{P} \cdot d\vec{s}$$



$$\text{Total charges} = \int_v \rho_p dv + \oint_S \vec{P} \cdot d\vec{s} = 0 \rightarrow \int_v \rho_p dv = -\oint_S \vec{P} \cdot d\vec{s}$$

$$-\oint_S \vec{P} \cdot d\vec{s} = \int_v \rho_p dv = -\int_v \vec{\nabla} \cdot \vec{P} dv$$

(divergence theorem)

$$\therefore \rho_p = -\vec{\nabla} \cdot \vec{P}$$

Divergence of P at a point in a dielectric gives the bound charge density of the point.

Material Response & Relative Permittivity

From Microscopic Polarization to ϵ_r

Engineering insight:

We do not measure microscopic bound charge directly.

We model material response using relative permittivity.

In this section, you will:

- Relate $\mathbf{P}$ to $\mathbf{E}$
- Understand the meaning of electric susceptibility
- Interpret the role of ϵ_r in Gauss' law

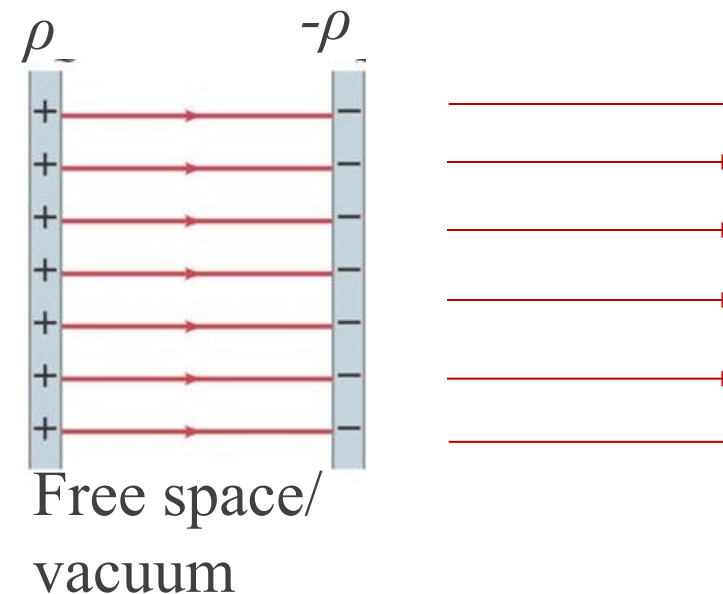
First postulate in material media

First postulate in vacuum or free space :

$$\vec{\nabla} \bullet \vec{E} = \frac{\rho}{\epsilon_0}$$

 (after polarization of a dielectric)

$$\vec{\nabla} \bullet \vec{E} = \frac{1}{\epsilon_0} (\rho + \rho_p)$$



**However, it is difficult to measure ρ_p , instead we use relative permittivity ϵ_r .

$$\vec{\nabla} \bullet \vec{E} = \frac{\rho}{\epsilon_0 \epsilon_r} = \frac{\rho}{\epsilon}$$

How ?

How relative permittivity ϵ_r relates to $\vec{P}$ and ρ_p ?

$$\vec{\nabla} \bullet \vec{E} = \frac{1}{\epsilon_0} (\rho + \rho_p), \text{ but } \rho_p = -\vec{\nabla} \bullet \vec{P}$$

Substitute and rearrange,

$$\therefore \vec{\nabla} \bullet \epsilon_0 \vec{E} = \rho - \vec{\nabla} \bullet \vec{P}$$

$$\vec{\nabla} \bullet (\epsilon_0 \vec{E} + \vec{P}) = \rho$$

If the dielectric medium is *linear and isotropic*, $\vec{P}$ is proportional to the applied electric field $\vec{E}$.

$$\vec{P} \propto \vec{E} \rightarrow \vec{P} = \chi_e \epsilon_0 \vec{E}$$

The proportionality constant χ_e is a dimensionless quantity called [electric susceptibility](#). It is difficult to measure quantity, hence seldom use in practice.

$$\vec{\nabla} \bullet (\epsilon_0 \vec{E} + \vec{P}) = \rho, \quad \text{replacing } \vec{P} = \epsilon_0 \chi_e \vec{E},$$

$$\vec{\nabla} \bullet \epsilon_0 (1 + \chi_e) \vec{E} = \rho$$

$$\epsilon_r = 1 + \chi_e \quad \text{The unitless } \epsilon_r \text{ is called } \textit{relative permittivity} \text{ or } \textit{dielectric constant}.$$

- ⚠ In exams, you may be asked to:
- Interpret ϵ_r and ϵ physically
 - Use it in derivations, not just restate it.

Substituting,

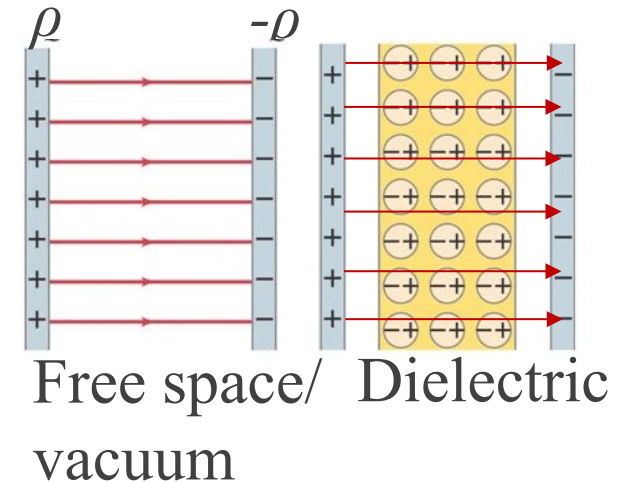
$$\vec{\nabla} \bullet \epsilon_0 \epsilon_r \vec{E} = \rho \rightarrow \vec{\nabla} \bullet \vec{E} = \frac{\rho}{\epsilon}$$

$\epsilon = \epsilon_0 \epsilon_r$ is called the *permittivity* of the medium. The SI unit of ϵ is F/m.

ϵ_r can be measured much more easily compared to χ_e , or ρ_p . ***

For example, $\mathbf{E}$ can be measured with and without a dielectric material in a unit surface area ($=1\text{m}^2$), and the ratio of these $\mathbf{E}$ s will inform the value ϵ_r . The factor by which $\mathbf{E}$ reduces in a dielectric relative to vacuum/free space is the ϵ_r

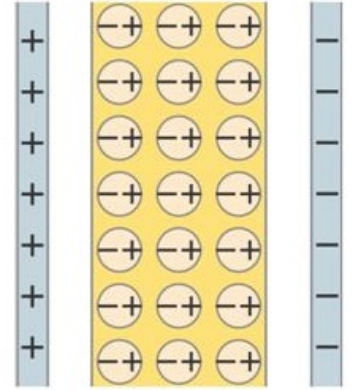
If ϵ_r increases, what happens to $\mathbf{E}$ for fixed ρ ?



How to find $\mathbf{P}$ from $\mathbf{E}$ & $\mathbf{D}$

$$\vec{\nabla} \cdot \vec{D} = \rho, \quad (1)$$

$$\vec{\nabla} \cdot \vec{E} = \frac{1}{\epsilon_0}(\rho + \rho_p) \rightarrow \vec{\nabla} \cdot (\epsilon_0 \vec{E} + \vec{P}) = \rho \quad (2)$$



From (1) and (2)

$$\therefore \epsilon_0 \vec{E} + \vec{P} = \vec{D} \rightarrow \vec{P} = \vec{D} - \epsilon_0 \vec{E}, \quad \vec{D} = \epsilon_0 \epsilon_r \vec{E}$$

For a given $\mathbf{D}$, $\mathbf{P}$ is a mean to reduce $\mathbf{E}$ in the dielectric materials.

Why will you want to reduce $\mathbf{E}$ in the dielectric?

Dielectric Breakdown & Applications

Dielectric Strength and Breakdown

Practical consequence:

Every insulating material has a maximum electric field it can withstand.

In this section, you will:

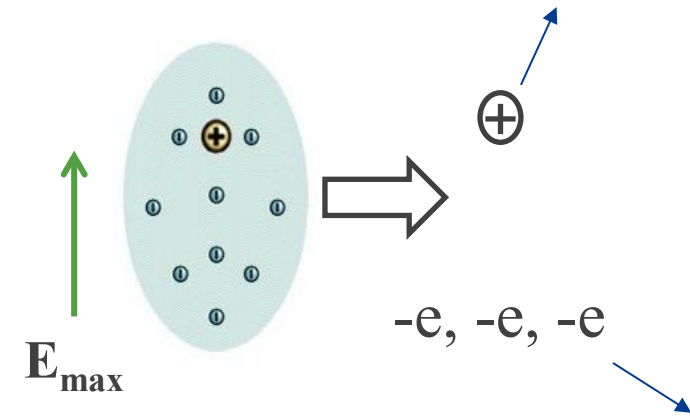
- Define dielectric strength
- Interpret material property tables
- Connect field strength to real engineering limits (e.g., corona, lightning)

Dielectric strength and breakdown

Properties of common materials

Material	ϵ_r	E_{max} , MV/m
Air	1.00054	0.8- 3
Water	78	-
Porcelain	6.5	4
Bakelite	4.8	12
Neoprene	6.9	12
Transformer Oil	4.5	12
Pyrex Glass	4.5	13
Paper	3.5	14
Polystyrene	2.5	20
Rubber	3.5	25
Polyethylene	2.5	50
Amber	2.7	90
Mica	2.4	200
Titanium Dioxide	100	6
Vacuum	1.000000	∞

The maximum electric field E_{max} that a dielectric material can withstand without *breakdown* is called Dielectric Strength.



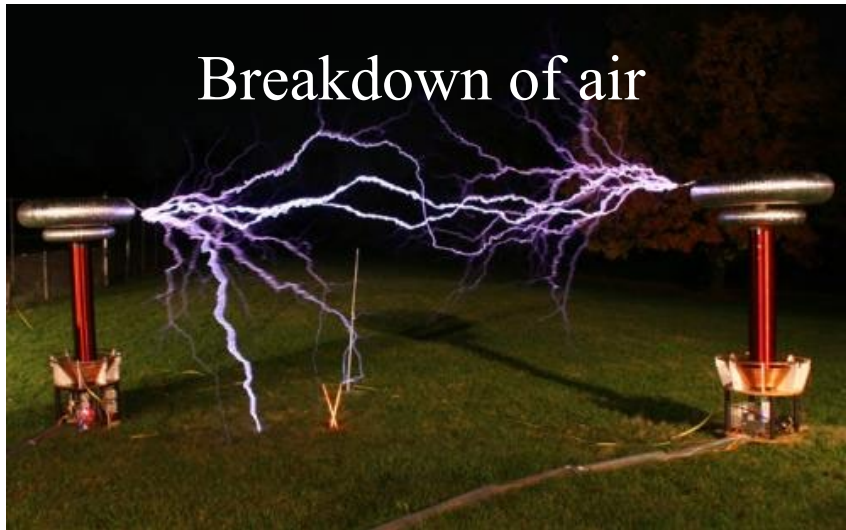
Dielectric breakdown

Observations:

- Air breaks down at ~3 MV/m
 - Mica extremely strong (200 MV/m)
 - Water high ϵ_r but low strength
- ⇒ High permittivity $\neq$ high breakdown strength.

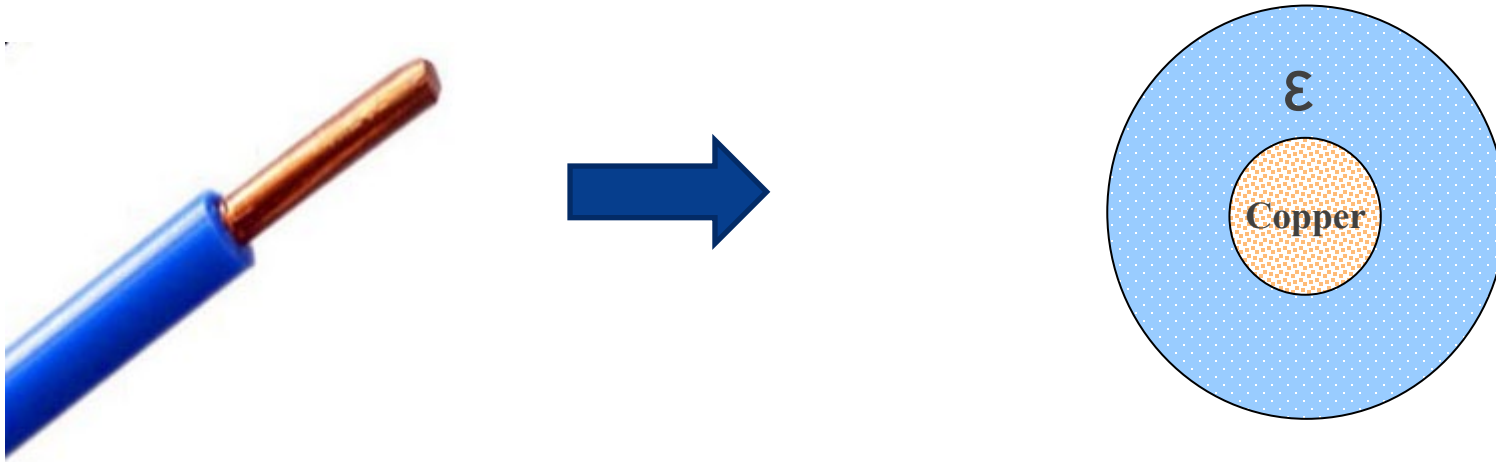
Dielectric breakdown of Air

Lightening, corona are examples of dielectric breakdown of air.



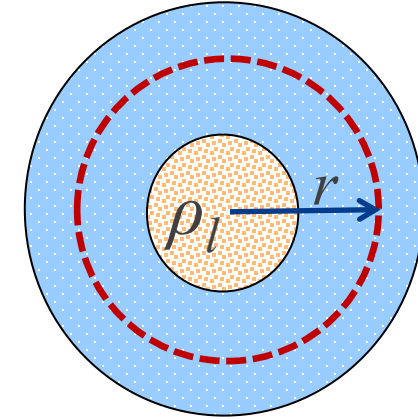
Example 4

A single-core power cable of circular cross section, as indicated in the figure, has a single insulation layer of the permittivity ϵ . The line charge density in the conductor is ρ_l [C/m]. Find the polarisation vector $\mathbf{P}$ of the insulating material. **Does $\mathbf{P}$ increase or decrease at the outer edge compared to the inner edge next to the copper conductor?**



Solution

Can you find $\mathbf{E}$ at radial distance r due to ρ_l ?



Find $\mathbf{P}$ in terms of $\mathbf{E}$ using $\vec{P} = \vec{D} - \epsilon_0 \vec{E}$, $\vec{D} = \epsilon \vec{E} = \epsilon_0 \epsilon_r \vec{E}$



Before You Go... (1 minute reflection)

This reflection will count toward optional bonus marks.



0 response submitted

What is the most important idea you are taking away from today's lecture? (Responses should reflect your ow...



Waiting for response...

Responses will be displayed as a word cloud

Wordcloud

All responses

<

1 of 3

>

Scan the QR or use link to join



<https://forms.office.com/r/vJs4eSWdQH>

 Copy link

Key takeaways from today

Inside conductor $\rightarrow \mathbf{E} = 0$

In dielectric $\rightarrow \mathbf{E}$ reduced but not zero

In vacuum $\rightarrow \mathbf{E}$ largest

Same Gauss law. Different material response.

Before Workshop 1:

Review Gauss's law examples and focus on *choosing the Gaussian surface*.

Re-derive the electric field and potential of a cylindrical conductor using Gauss's law.

Review why $E = 0$ inside a conductor. Review the dielectric strength and breakdown.

Workshops will focus on applying these ideas, not reintroducing theory.

Review and further practice (Recommended)

Practice Example 3-6, 3-8, 3-11, 3-12, 3-13 from the textbook by D.K Cheng
We will practice several other examples in Workshop on Friday.

Challenge question:

Check out example 3-3 and then attempt Problem P.3-2 (a) in page 145 of the textbook by D.K Cheng.

Ans:

$$V_{max} = \frac{m}{e} \left(\frac{u_0 h}{w} \right)^2$$

